

MP1: Orientation and Weekend Entertainment

Due Time: Refer to the course web-site

Objective

In this programming assignment, we familiarize with the learning tool, Unity, that we will be using for the rest of this quarter. You should also use this opportunity to:

- Evaluate how comfortable you are with the kinds of learning that is expected: *on how to use Unity as a tool*. Note, this is very different from “*learning how to use Unity to build a game*”, here, we want to learn as little as possible to accomplish what we need.
- Ensure that you understand exactly what is involved in submitting a programming assignment.

This assignment should take you no more than 5 or 6 hours to complete. Most of the time should be spent on asking and looking for answers to: “*how do I make Unity do ...*”

Approach

Please refer to [my attempt at the solution](#) (WebGL build, simply click and run). Initially, your world consists of a **CreationPlane** and a **CreationTarget**:

- **CreationPlane**: is a **Quad** with
 - Transform:
 - Position: 0, 0, 0
 - Rotation: 90, 0, 0 (such that the Quad is on the x/z plane)
 - Scale: 14, 14, 14
 - Note: the corners of the plane spans between -7 to $+7$
 - If you choose to work with a **Plane**, your transform settings will be different, but, it is important to make sure the size and location of the CreationPlane to be specified as above.
- **CreationTarget**: is a sphere with
 - Transform:
 - Position: user defined!
 - Rotation: 0, 0, 0
 - Scale: ~~0.1, 0.1, 0.1~~ 0.5, 0.5, 0.5
 - Color: black

You must also support a radio button system that allows the creation of a Sphere, Cylinder, or Cube. Here are the specifications for each of the shape:

- Sphere: (not circular so we can observe the rotation)
 - Scale= (0.5, 1, 1)
 - Speed = 1 unit **per second**, along the y-axis
 - Rotation about the y-axis: 45-degree per second
 - Range: $0 < y < 5$
 - Color:
 - When traveling in the positive-Y: color= (1, 1, 1)
 - When traveling in the negative-Y: color=(1, 0, 1)
- Cylinder: (not symmetrical so we can observe the rotation)
 - Scale = (0.5, 1, 1)

- Speed = 1.5 unit **per second**, along the z-axis
- Rotation speed = about y-axis, 90-degrees per second
- Range: $0 < z < 5$
- Color:
 - When traveling in the positive-Z: color=(1, 1, 1)
 - When traveling in the negative-Z: color=(0, 1, 1)
- Cube:
 - Size of 1 (scale = 1)
 - Speed = 2 unit **per second**, along the x-axis
 - Rotation speed = about y-axis, 135-degrees per second
 - Range: $0 < x < 5$
 - Color:
 - When traveling in the positive-X: color=(1, 1, 1)
 - When traveling in the negative-X: color=(1, 1, 0)

Behaviors

There are three basic behaviors that you must support.

- 1) **Primitive Created:** This is a radio button where **at most** one of the check-marks can be checked.
 - a) When user click on a primitive type, a new primitive with the corresponding type is created at the **CreationTarget** and the check mark stays.
 - b) Clicking on the same primitive type consecutively the second time will un-check and not create a new primitive.
 - c) When user click to delete a primitive, all check marks must be off.
 - d) In all cases, a newly created shape should be resting on the **CreationPlane** and attempt to travel according to the above definition. If the creation position is larger than the valid range of 5, the created shape will then travel in the negative direction. After a shape's first change of traveling direction, it should never cross the 0 to 5 range.
- 2) **Creation target manipulation:** You must support Left-Mouse-Button click (LMB):
 - a) When LMB is over the **CreationPlane** and **not** over any other objects, the **CreationTarget** is moved to the mouse click position.
 - b) **Note:** CreationTarget must always be entirely above the CreationPlane.
- 3) **Object deletion:** LMB over a created object should delete the object. **Note:** one can never delete the **CreationPlane** or the **CreationTarget**.

Hints

1. My MainCamera is located at (0, 8, -16), with rotation (23, 0, 0). FOV of 45.
 - Learn the trick: in the Hierarchy Window, select the MainCamera, RMB→AlignWithView and see what happens!
2. For radio buttons, I used [Toggle](#) and [ToggleGroup](#). Creating ToggleGroup is slightly tricky, you will select the GameObject, and then, Inspector, click on the Add Component →UI.
3. I googled “unity elapsed time” to try to find out how to work with **per-second** movements and rotations. Take a look at **deltaTime** (<https://docs.unity3d.com/ScriptReference/Time-deltaTime.html>),
4. To avoid mouse-click selecting the **CreationTarget**, I place the corresponding sphere into a different **Layer**, and **Raycast(ray, hitInfo, layer-1)** [the default layer] to click select floor or the created objects.
 - If mouse click is on the floor, I move the CreationTarget over to the (x, someValue, z) where x, and z are from the mouse-click.
 - Do a google on “unity layer” to find out more about Layers.
5. Take a look at **Rotate(Axis, degree)** to rotate a GameObject

Credit Distribution

1. (5%) Initial frame: **CreationPlane** and **CreationTarget**
 - (5%) Correct size/color for both
 - (5%) Both entirely visible from the runtime camera
 - (-50%) **If the CreatePlane is not entire visible at 1920x1080**
2. (20%) **CreationTarget** interaction
 - (10%) Proper placing of **CreationTarget** at LMB (place anywhere on the **CreationPlane**)
 - (5%) The sphere drawn always above **CreationPlane**
 - (10%) Can never delete **CreationTarget**
3. (20%) Primitive Created radio buttons support
 - (5%) Initially none checked
 - (5%) Only create when set to check (unchecking does not create)
 - (10%) Click to check to create a primitive at **CreationTarget**
 - (10%) None checked after deleting a primitive
4. (25%) Created Primitive Appearance and Behavior
 - (5%) Create at the **CreationTarget** position with entire primitive above the **CreationPlane**
 - (5%) Proper color changes: white when move in positive direction
 - (15%) Movements Speed + Rotation: Sphere:Y/slow, Cylinder:Z/faster, Cube: Z/fastest
 - (10%) Movement range: always between 0 to 5
5. (15%) Click to delete:
 - (10%) Proper deletion of mouse clicked primitive
 - (10%) Delete does not move **CreationTarget**
 - (10%) Cannot delete the **CreationPlane** or **CreationTarget**
6. (10% **CSS551 only, Extra credit for CSS451**) Support for Sliders
 - Include 3x slider bars controlling the RGB channels of the **CreationTarget**
7. (15%) Submission and Others
 - (10%) Zipped all source code + exe folder
 - (10%) Proper zipfile naming: FirstLastNameMP1.zip
 - (10%) EXE Runs as a re-sizable window (NOT full screen)

Submission:

- **Source code:** just the one folder: **Assets**
- **Executable:** Include a separate **EXE** folder, and build your executable into this folder.
 - **Build** → Player Settings
 - **Fullscreen Mode:** Windowed
 - **Resolution:** 1920x1080 is fine
 - **Resizable Window:** check this!

Grade:

- **CSS551 students:** Your grade will be based on 110 converted to percentage
- **CSS451 students:** Your grade will be based on the 100, if you implemented the slider bars, the 10-point will go towards your extra credits for this class

Extra Credit:

You can do 100 million (and more) different things to test/learn/demonstrate your knowledge of Unity. I will give out up to 10-points of extra credits on any experiment/features you implement (must be working)!

Extra credit ideas: it is **NOT** as interesting to spend time making your mp “*beautiful*” or do what we can do but just more, e.g.,

- Less interesting to allow the creation of more primitive types (more of the same), we already know how to do this
 - Instead: can you create a car? [something different]
- Less interesting to place a nice looking geometry for **CreationTarget** (again, we know how to do this)
 - Instead: support multiple **CreationPlane** in different dimensions (make sure I know how to test the basic functionality)
- Other interesting things:
 - Global behavior control: click to stop all movements
 - Individual object control: stop/start individual objects [e.g., with an extra deletion button], or customizable movements
- You can experiment with Unity’s Scene support and create separate scenes for your different test cases and extra stuff

As I said, there are 100 million different things you can do.

WATCH OUT: This is due on next week. My plan is to grade your submission and return them to you on following week so that you know what is expected of you!! MP2 is coming on next week. Start early, DO NOT fall behind, it will be difficult to catch up.